

Searching an ordered table:

If the table is stored in ascending or descending order of the record keys, there are several techniques that can be used to improve the efficiency of searching. This is especially true if the table is of fixed size.

In an unordered table, in order to detect the missing key, we will require n comparisons. But we can easily find the missing key in an ascending order table of keys.

If there are large number of records, large number of requests to search the records. It is very difficult or slow process to search the entire records. So if the records are sorted in ascending order, then there may be only fewer comparisons. This will increase the efficiency of searching.

Indexed Sequential Search:

This one of the techniques used to search a record in a sorted file. But it involves an increase in the amount of space required.

An auxiliary table, called an Index, is set aside in addition to the sorted file. Each element in the index consists of a key K_{index} and a pointer to the record in the file that corresponds to K_{index} . The element in the index, as well as the elements in the file, must be stored on the key. If the index is one eighth the size of the file, every eighth record of the file is represented in the index.

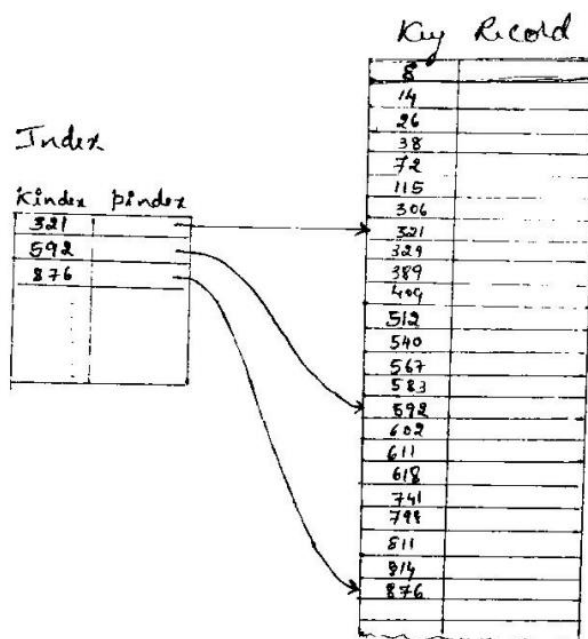


fig: Indexed Sequential file

The real advantage of the indexed sequential method is that the item in the indexed table can be examined sequentially if all the records in the file must be accessed, yet the search time for a particular item is sharply reduced. A sequential search is performed on the smaller index rather than on the larger table. Once the correct index position has been found, a second sequential search is performed on a small portion of the record table.

If the table is so large that even the use of an index does not achieve sufficient efficiency, a secondary index can be used. The secondary index acts as an index to the primary index, which points to entries in the sequential table. This is illustrated in the following figure.

